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Frontier and Localhost: A Survey of How Production AI Learns Outside the Weights

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ABSTRACT Production AI now learns mostly outside the model weights, and the industry has built, piece by piece, most of an optimizer for that layer without ever assembling it. After a failure, teams edit instructions, skills, memories, tools, orchestration graphs, evaluation suites, and governance rules far faster than any weight update, and two earlier results of this series suggest why that is the right place: errors concentrate in a few key decisions that fail in a small, recurring catalogue of modes. This survey of 256 production and research systems reads that layer as one optimizer with two halves. The first traces a recurring failure to one scaffold coordinate, proposes one edit, and keeps it only if a gate shows the deployment's own failure rate falling. The second decides how far the edit may travel. What a deployment can automate is fixed by the feedback its gate receives: an exact check, an observed outcome, a judgment, or a delayed reward. An accepted edit improves only what that gate measured. Across the corpus the first half is routine and much of the second exists in parts: evidence is batched before acceptance, retirement is gated, lineage ships in products, formats and registries have converged, and the first federations of deployments exist. No surveyed system assembles those parts into promotion across organizations under versioned lineage, access control, and rollback. We give the four-property criterion that would, the failure surface left open until it exists, and, for each signal class, the piece that is missing.

INDEX TERMS AI governance, agent memory, agent skills, evaluation, large language models, LLM agents, prompt optimization, self-improving systems, survey

I. INTRODUCTION

The standard mental model of a system built on a large language model (LLM) is the model. Pre-training fixes the weights, fine-tuning nudges them, and deployment wraps a thin layer of prompts and tools around the result. Production in 2025 and 2026 has stopped behaving this way. Frontier weights change on a cycle of months; the systems built on them change daily, in instruction files checked into repositories, memories that persist across sessions, skills loaded on demand, registered tools, evaluation suites that block a merge, agent topologies, and pull-request gates on shared rule repositories. A growing share of the adaptation that makes a deployed system reliable now happens outside the weights, in a fast-changing local scaffold wrapped around a slow-changing frontier model.

Read one at a time, these artifacts look like unrelated engineering. A skill file is a procedure the model did not have; a repository rule is a convention for one project; a retrieval connector is memory access for one workflow; a tool registry provisions a capability the base model cannot reach; a pull-request review against a shared rule repository is a promotion gate. Read together, they are one adaptation surface, the non-model part of what Zaharia et al. call a compound AI system [1]. This survey asks what optimization discipline that surface needs, and finds that the field has built the layer and, as far as public evidence shows, not the optimizer that should govern it. Most deployments maintain the layer as patchwork: a failure happens, someone adds a rule, the pile grows, nobody knows which addition helped, and fixes leak into contexts that never validated them.

The disciplined alternative is one loop with two halves. Its first half turns a recurring failure into one tested edit to the scaffold and keeps the edit only if the deployment's own failure rate falls; the second decides how far the edit may travel, to a team, to an organization, or to every customer, and lets it travel only as far as evidence supports. Set side by side, the 256 systems in this survey tell one story about that loop. The first half is routine: in most research systems and a growing number of production ones it closes under an automatic gate. Much of the second half exists in parts: the newer systems batch evidence before they accept an edit, retire artifacts under a gate, ship memory with its lineage, converge on one artifact format and one tool layer, and in a few cases federate edits across deployments inside one operator. None of the 256 assembles those parts into a second half governed by versioned lineage, access control, and rollback. The reason is structural rather than a matter of effort. What separates the two halves is the feedback the gate receives: a test or an observed outcome can gate an edit automatically, a judgment only once the judge is calibrated against people, a reward that arrives months later not edit by edit at all. Because an accepted edit is an improvement only in what its gate measured, a lift earned in one deployment does not travel to another unless the receiver measures the same thing or measures again.

This article makes five contributions.

1. The loop that turns patchwork into optimization, in plain language and as a gated update rule, with the five conditions under which it is an optimizer and the two things those conditions imply: the measure bounds the lift, and the loop optimizes rather than evolves (Section III).
2. Four classes of feedback signal, from an exact check to a delayed reward, each fixing what the gate can be and which failure dominates (Section IV).
3. A synthesis of 256 production and research systems, scored on six substrates against a published rubric, with the scores released, and read as six trends, five missing pieces, and four ways of building the loop (Sections V and VI).
4. A design space indexed by the gate on each half of the loop, and the convergence of five research families on the same loop (Sections VII and VIII).
5. A four-property audit criterion for governed cross-organization scaffold optimization, which no surveyed system meets, with the failure surface its absence leaves open and the open problem each signal class poses (Sections IX to XI).

The survey runs no experiments. Its evidence is the published behavior of 256 systems, scored on a released table.

II. WHY ADAPTATION LIVES OUTSIDE THE WEIGHTS

A. BREADTH IN THE WEIGHTS, DEPTH IN THE SCAFFOLD

Frontier training optimizes for breadth. The weights are tuned to generalize across millions of unseen deployments, and the objective smooths over exactly what one deployment cares about: local naming conventions, contradictory preferences between customers, rare workflow-specific failures. Reinforcement learning from human feedback reduces output diversity and homogenizes preferred style across users [2], [3], and instruction-tuning data rewards patterns that generalize across the annotator pool, so the frontier model is trained to disfavor one customer's idiosyncrasies. Tiwari et al. [4] reach the same place from the optimization side: in-context adaptation is cheap but capacity-bounded, in-weights adaptation is expressive but induces forgetting and loss of plasticity [5], so a system that learns slowly in the weights needs a cheap outer layer that learns fast outside them.

Production reliability is won in depth, inside one deployment, and it turns on what global training had to smooth away. Encoding every deployment in the weights is ruled out three times over: fine-tuning per customer multiplies the training, evaluation, and serving lifecycle by the number of deployments; weights ship quarterly at best, while the deployment wants daily updates; and overfitting the weights to one customer degrades the neighbors whose preferences contradict. In principle, per-customer adapters [6], distilled models, and learned routing could absorb some of the residual, the failures that remain after training. In practice they are rare. Adaptation lands in the scaffold, the one substrate that is cheap, inspectable, reversible, scoped to one customer, and updatable daily.

B. WHY THE SCAFFOLD IS THE RIGHT ARCHITECTURE: THE TWO EARLIER RESULTS

The case for learning outside the weights does not rest on cost alone. The two papers that precede this one in the series supply the hypothesis that explains the pattern the corpus shows, and each states the scope on which it rests. *Beyond Exponential Decay* [7] argues that long outputs do not fail at the rate a uniform per-token error model predicts, because reliability hinges on a sparse set of key decisions. Its evidence is twofold: a direct measurement of long-range dependence on one natural-text corpus, where under a tenth of tokens depend on distant context, and the accuracy that majority voting recovers on reasoning benchmarks. The argument stops where the number of decisions grows with the task, as in long-horizon agentic work. For a deployment that operates inside a bounded slice of recurring tasks, tools, and evaluator expectations, *The Architecture of Errors* [8] argues that those decisions fail in a small recurring catalogue of modes whose size grows only slowly with the number of observed failures. A targeted intervention budget that grows polylogarithmically in sequence length, and becomes constant once the catalogue saturates, then covers the residual. The paper's own negative result marks the limit: no finite catalogue covers an unbounded domain.

If those results hold in general, a deployment's errors are few enough to fix one at a time, and a small catalogue of fixable modes is exactly what a cheap, inspectable, reversible artifact layer is for. This article asks where the fixes live, and the corpus answers: in persistent artifacts outside the weights, one recurring mode at a time, which is the pattern the

hypothesis predicts. On that hypothesis the gated and scoped layer is not a workaround for an inadequate model. It is the architecture the error structure calls for: a frontier model that generalizes, a local scaffold that specializes, and reliability that comes from governing the specialization. The survey's own findings, in Sections VI to IX, stand without it.

C. RELATION TO EXISTING SURVEYS

The neighboring surveys explain how agents evolve, what their memories store, how context is engineered, how skills are evaluated, and how harnesses are built. None asks the question this one is organized around: who decides that an edit is good, and how far may the edit travel. That question has four parts: whether a feedback loop closes at all, whether its gate is a program or a person, what the gate can observe, and whether promotion beyond one deployment is governed. Table 1 says, for each neighbor, what it classifies by and where this survey departs from it. Two neighbors come close on the governance part. Ding et al.'s survey of persistent-state agents finds the literature concentrated on accumulating and retrieving state rather than on governing, recovering, or relinquishing it [9], and the memory-security survey of Lin et al. scopes and rolls back memory for safety [10]; neither asks who gates an edit, and neither covers promotion across organizations.

A reader will also expect the prompt-optimization lineage: OPRO, EvoPrompt, Promptbreeder, DSPy, and TextGrad [18]–[22]. That lineage optimizes a prompt or program once against a development set and is not a deployed loop; agentic context engineering [23] does persist its artifact, an evolving playbook, but never promotes it. The closest precedents optimize the scaffold itself under a benchmark score: agent symbolic learning updates prompts, tools, and pipeline as symbolic parameters from textual feedback [24], STOP's scaffolding program improves its own improvement code [25], and a coding agent edits its own harness to raise its score [26]. What changes here is that the edited artifact persists in a deployment and must clear a second gate before it earns any scope beyond it (Supplementary Material S7).

III. THE LOOP: ARTIFACT-LAYER DESCENT

A. THE LOOP IN PLAIN LANGUAGE

We call the disciplined version of scaffold maintenance artifact-layer descent. It is one loop with two halves, run again and again over the life of a deployment: Loop 1 fixes a recurring failure inside one deployment, and Loop 2 decides how far the fix may travel. Fig. 1 draws the first half. The deployment runs and logs what goes wrong: failed outputs, user corrections, rejected answers, evaluation regressions, tool-call errors. The loop does not react to single failures. It clusters the log into recurring modes and takes up the ones that matter most, ranked by how often each recurs and how badly it fails. For such a mode it locates the scaffold coordinate responsible, which may be a stale memory entry, an under-specified instruction, a missing tool argument, a brittle retrieval rule, or a misrouted agent edge, and it proposes one candidate edit to that coordinate.

Then it measures. A deployment serves one **patch**: the slice of users, tasks, and conventions it handles. The failure rate on that patch's recurring tasks is the **patch loss**. A gate accepts the edit if the patch loss falls and rejects it otherwise. An accepted edit stays in the scaffold, versioned; one that a later check shows made things worse is rolled back. That is the whole of Loop 1, and it is what most research systems and a growing number of production systems now do.

Loop 2 is the step most systems skip. An engineer notices that a coding agent keeps misnaming a class in one repository. The fix is a one-line rule in that repository's instruction file. That same rule could be promoted to a team-wide template, to the company's core instructions, or to a skill registry shared across companies. The edit did not change; its scope did. And the gate that should let it through gets stricter at each step, because the loss it is tested against is now an average over more deployments, and a fix that helps one repository can hurt another.

The scopes nest (Fig. 2). USER is one person's account or fork; PROJECT is one repository, workflow, or product workspace; STACK is a curated bundle of projects under one team; a **tenant** is one organization or customer; CORE is the scope every tenant shares. Each level's loss is the average failure rate over the deployments it contains. Promotion that reaches CORE crosses tenants, which is what cross-tenant promotion means here. Loop 2 promotes an edit upward only as far as the evidence reaches; we call that limit the edit's **evidence radius**. The check that belongs at each rung is **receiving-side evaluation**: a check run where the edit arrives, not where it was made. In optimizer terms, Loop 1 sets the step (how much one accepted edit changes the local scaffold) and Loop 2 sets the target (which average failure rate the scaffold is tuned to reduce). Governance is therefore not administrative overhead. It is the regularization that keeps a small fix from being aimed at the wrong target.

B. WHEN THE LOOP IS AN OPTIMIZER

The loop is an optimization process, rather than a description of good practice, when five conditions hold:

1. a measurable patch loss, such as a failure rate, a correction rate, or an evaluation score;
2. captured residuals, so that failures are logged rather than lost;
3. credit assignment, so that a failure can be traced, at least sometimes, to the scaffold coordinate that caused it;
4. a synthesized candidate edit to that coordinate; and
5. a gate that checks whether the edit lowers the loss before the edit is kept or promoted.

The conditions presuppose frozen frontier weights, edits that persist as inspectable artifacts rather than transient prompts, and promotion that widens an artifact's scope under a gate rather than averaging comparable updates as weight-space federated learning does.

TABLE 1. Neighboring surveys and where this one departs from each. This survey classifies by loop closure, gate type, signal class, and governance, and is the only one to cover promotion across organizations.

Survey	Scope	How this survey differs
Gao et al. [11]	self-evolving agents	organizes by what, when, how, and where an agent evolves; the gate that accepts an edit, and the edit's scope, are mentioned but not classified
Fang et al. [12]	self-evolving agents	organizes by inputs, agent system, environment, and optimizer; loop closure appears in passing, gates and promotion do not
Zhou et al. [13]	self-evolving coding agents (108 methods)	organizes by evolution target, timing, and code signal; closest on reversibility, with no gate or promotion axis
Ding et al. [9]	persistent-state agents (435 works)	organizes by authority, scope, mutability, provenance, and recoverability of state; governance is an axis, the gate on an edit is not
Lin et al. [10]	memory security (~90 works)	organizes by lifecycle phase and security objective; principal scoping and rollback are axes, feedback and promotion are not
Guo et al. [14]	agent system and harness design	organizes by six runtime responsibilities; touches all four questions without classifying by any
Xia et al. [15]	skill-mediated agents (8 systems)	organizes by four responsibility layers and ten patterns; loop closure and governance appear as patterns, not as axes
Ding et al. [16]	skill evaluation and evolution	organizes by four evolution paradigms and six benchmark classes; evaluation is the subject, the gate it feeds is not
Mei et al. [17]	context engineering (1,411 papers)	organizes by components and system implementations; adaptation loops lie outside its scope

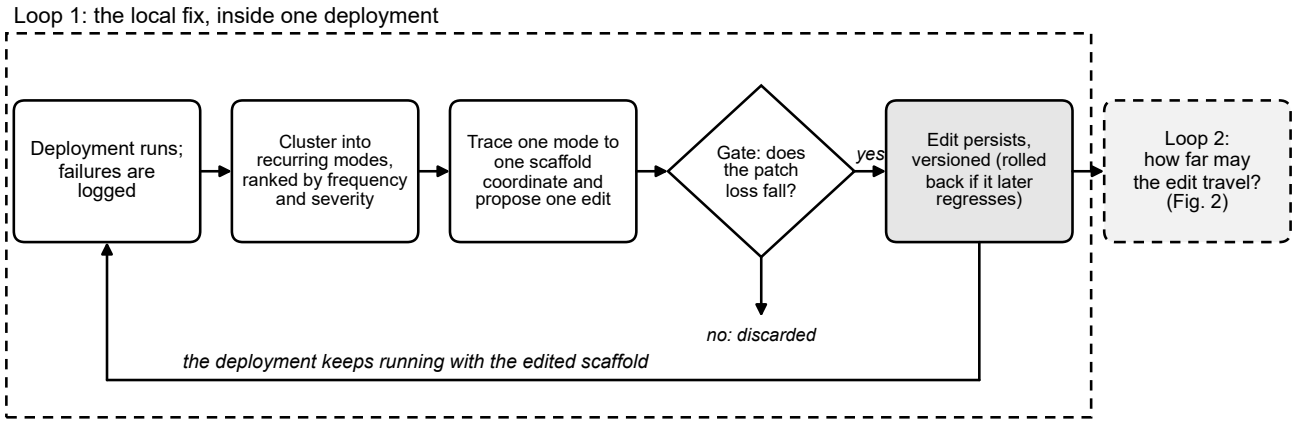


FIGURE 1. Loop 1, the local half of artifact-layer descent. A recurring failure is traced to one scaffold coordinate, one edit is proposed, and a gate keeps the edit only if the deployment's patch loss falls. An accepted edit then enters Loop 2 (Fig. 2).

Written out, the patch loss of deployment D is

$$L_D(\theta_s) = \mathbb{E}_{x\sim D}[\ell(\pi(\theta_M, \theta_s, x))]\,,$$

where θ_M are the frontier weights, fixed during deployment, θ_s is the scaffold, a structured object over the six substrate coordinates of Section V, and ℓ is an operational loss. A failure yields no analytic gradient. It yields a candidate edit z_t, proposed by reflecting over the accumulated failure history. The gate accepts the edit when its noisy estimate of the change in loss clears a margin,

$$G_t(z_t) = 1 \iff \widehat{\Delta}_{z_t} L_D(\theta_s^t) + \lambda\,\Omega(z_t) \le -\tau_t,$$

where $\Omega \ge 0$ penalizes the complexity of the edit and the breadth of its promotion, and $\tau_t \ge 0$ is the required margin. Production systems implement this inequality without ever writing it down: an evaluation blocks a merge on regression, a reviewer rejects a vague rule, a governance system holds an artifact back until evidence accumulates. Under a bounded-bias assumption on the gate, the update satisfies a one-step descent inequality. When that bias is summable over the run, the loss converges and improving edits stop clearing the gate; the scaffold is then gate-stable, which is a statement about the gate, not about optimality (Appendix XII). Where a condition fails, the picture is an analogy, and the survey marks which systems meet which conditions.

Each half of the loop has a gate, and we name three kinds by who decides. A gate is **Auto** when a program applies a check before the edit goes through: an evaluation score, a verifier, a reinforcement-learning threshold, or a language-model judge [27] with a quantified agreement rate. It is **Human** when

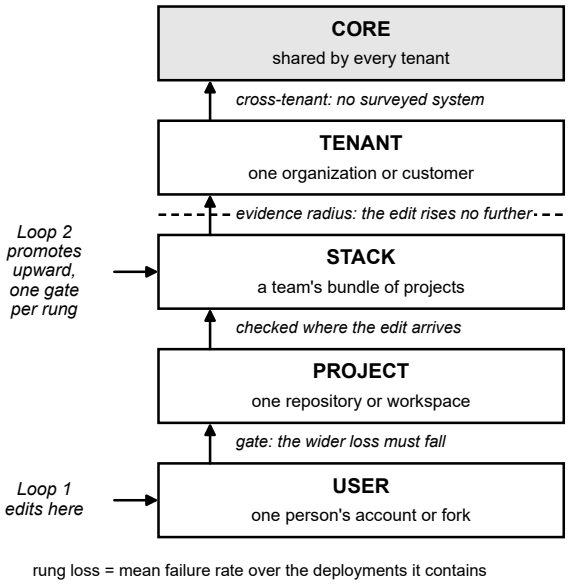


FIGURE 2. Loop 2, the promotion half. Scopes nest from USER to CORE, each rung's loss averages the deployments it contains, and an edit rises one rung at a time, only while a check at the receiving side shows the wider loss falling and only as far as its evidence radius. Promotion into CORE crosses tenants.

a person decides. It is **None** when nothing decides, either because that half of the loop is absent or because the step runs unchecked, as when a system pushes every edit to its neighbors without a test. Who writes the candidate is recorded separately and is not a gate: the author may be an agent reflecting over failures, an engineer at a desk, or a contributor registering an artifact. We write a system's two gates as a pair, Loop 1 first: [Auto $\times$ Human] means an algorithmic local gate and a human promotion gate. These pairs index the design space of Section VII.

Condition 1 says the loss must be measurable. It does not say what measures it, and that turns out to be the question that sorts the corpus.

C. FROM PATCHWORK TO DESCENT

Three stages of scaffold maintenance recur across the corpus, and the five conditions separate them. Take the coding agent above that keeps misnaming a class. At the first stage, patchwork, an engineer adds a rule to the instruction file after noticing the failure; nobody measures whether it helped, nothing is promoted on evidence, and the file sprawls with every incident until nobody knows which rule does what. At the second stage, random artifact search, the system tries many variants of the file under the test suite and keeps the best. The suite is the only signal and no failure is traced to the rule that caused it; the winning variant has learned the suite as much as the task. At the third stage, artifact-layer descent, the recurring failure is traced to the one rule responsible, one edit is proposed, the gate checks that the patch loss falls, and promotion stops at the evidence radius and can be reversed. A loop reaches the third stage once the five conditions hold, and its dominant failure becomes evaluation fragility, because the suite that gates the edits is itself a scaffold artifact that drifts and can be gamed (Section X). Production deployments that reach it on Loop 1 then promote in one of two ways: behind a human gate, as Spotify [28] and SkillForge do, or, in Letta's Context Repositories alone, by an ungated push that carries lineage but no evaluation.

Two consequences of condition 1 follow the loop everywhere, and the survey keeps returning to them. The first is that the measure bounds the lift. An accepted edit is an improvement in what the gate measured, on the tasks the gate sampled, and nothing more; a rule that lowers the failure rate on one repository's suite has earned nothing about another repository's. Every loss has a scope. That is why Loop 2 cannot carry Loop 1's evidence: the receiving scope measures a different mix of tasks, so a promoted edit either travels under a measure the scopes share or re-establishes its lift where it arrives, which is what receiving-side evaluation is for. Where tenants measure different things, a lift does not transfer at all without it.

The second is that the loop optimizes and does not evolve. It needs a defined measure before it can start and descends toward that measure's ceiling; at the ceiling no edit clears the gate, which is what gate stability means. It does not discover a better goal than the one its measure encodes. Two scaffolds optimized for the same goal under the same measure need not converge to one version; each stops at the ceiling by its own route. A system that edits its own measure, the reflexive gate of Section IV, moves the destination rather than evolving toward an unknown truth. The distinction matters because the corpus is often described in the vocabulary of evolution; what the gated systems in it actually do is descent.

IV. WHERE THE SIGNAL COMES FROM: FOUR CLASSES OF FEEDBACK

Four scaffolds all run the loop of Section III. A coding agent changes one line in its repository rule file; the test suite runs and reports pass or fail within seconds. A scheduling assistant changes how it phrases a booking request; the appointment either appears on the calendar or it does not, and the system learns which an hour later. A marketing assistant changes the tone rule in its instructions; nobody can say whether the next draft is better, only whether a reviewer prefers it. A research assistant changes how it chooses which direction to pursue; whether the choice was good becomes visible months later, after many other edits have landed. All four produce a candidate edit. They differ in what the gate can know about it.

That difference is the whole of this section. The gate needs an estimate of one quantity: whether the edit lowered the patch loss. That estimate comes from a feedback signal. Signals differ in three ways: how soon they arrive, how exact they are, and whether there is a fact of the matter at all. Those three differences sort every surveyed system into four classes (Table 2), and the class fixes what the gate can be, which

half of the loop can close automatically, and which failure dominates.

The classes are a partition, not three independent axes: a system is assigned by the property that binds its gate first. Delay binds before anything else, so an outcome that arrives months later is C4 even when it is objective; the absence of ground truth binds next, so an immediate judgment is C3; and among signals that are prompt and objective, exactness separates C1 from C2.

C1, exact. A deterministic checker returns pass or fail, or an exact score, immediately and repeatably: unit tests, compilers, type checkers, exact-match answer keys, proof checkers, schema validators. DGM accepts a self-modification only after it passes in a sandbox on coding benchmarks [29]; AlphaEvolve scores candidate programs with a cascade of exact evaluators [30]; SkillOpt gates a skill edit on a held-out exact-match split of a math benchmark [31]. Because the check is exact for the artifact and the test it runs, and repeating it returns the same answer, one run settles the checked case. What the check does not settle is the patch loss: the suite is a sample of the patch's task mix, and the agent that runs under the edited scaffold is stochastic, so the gate's estimate of the loss change still carries sampling error. The gate can be a program and its margin can be small, but the margin absorbs sampling error, not error in the check. The improvement the check certifies is also bounded and specific: an improvement in what the check measures, on the cases it ran, not an improvement in general. Two things stay hard. A failing test localizes the output, not the scaffold coordinate that caused it, so credit assignment remains open; and a checker that is also the training target can be gamed. The dominant failures are test overfitting and scaffold bloat, because cheap acceptances accumulate.

C2, observable outcome. The environment or the user emits an outcome after an episode: a booked appointment, a closed ticket, a merged pull request, a tool call that returns without error, a retrieved document that contains the answer. The signal is real but one episode late, noisy, and confounded, since a user can accept a wrong answer and reject a right one. Gates of this class include Devin's merged pull requests [32], SAGE's task success on AppWorld [33], SkillRL's success-rate threshold [34], and PACE's failure signals from a deployed small model [35]. The gate can still be a program (a reinforcement-learning threshold or a held-out task suite), but it needs a positive margin, and it absorbs the noise by batching episodes or by requiring agreement across samples (Appendix XIII). Credit assignment takes its most stubborn form here: the outcome says *that* the episode failed, not *why*. Most production deployments live in this class, and its signal arrives through a channel an attacker can control, which is why poisoned memory and tool-output injection dominate.

C3, judged. There is no fact of the matter. An essay, a marketing page, a design, a pitch deck, or a code review for style can be scored only by a judge: a human rater, a language-model judge, a rubric, or a comparison against references. SkillForge accepts a skill diff when its judge agrees with human engineers more than nine times in ten [36]; CASCADE consolidates memory under a human-and-agent gate [37]; Anthropic's skills repository promotes a skill when a reviewer approves the pull request. The gate can be a program only after the judge has been calibrated against people, because the judge's bias is now the estimator's bias. Residuals, meanwhile, are captured only when someone objects, so condition 2 depends on complaint channels rather than instrumentation. The dominant failure is Goodhart's law in the form measured for reward models: optimizing a proxy past a point lowers the true score [38]. Once the judge is the target, the scaffold learns the judge. Nor is a human judge stable: Shankar et al. find that evaluators' criteria drift as they grade more outputs [39].

C4, delayed and sparse. The outcome arrives after weeks or months, once per many edits, in a world that has moved on. Research programs, product strategy, and open-ended discovery are of this kind; the clearest instances are automated research pipelines such as the AI Scientist, which generates ideas, runs experiments, and writes the paper [40]. NanoResearch merges candidate skills by semantic compatibility rather than by a measured loss change [41], and CORAL promotes findings into shared memory on a heartbeat rather than on an evaluation [42]. AlphaEvolve's evolutionary database can look like an instance, and is not one: it balances improving the best programs against diversity only after every candidate has been scored by its exact evaluators, so it is a search strategy inside the exact class, not a gate for a delayed signal [30]. With no per-edit estimate to gate on, the gate degenerates to proposal pruning plus proxy milestones, and the descent argument covers only the composition of pruning and whatever gate exists at the horizon (Appendix XIII). Temporal credit assignment across many edits is open. The dominant failure is drift: the scaffold ages faster than the signal that would correct it. An optimizer in this class also cannot be told from random search by its outcomes alone; Ong et al. [43] report the same indistinguishability for harness optimizers.

The judge is part of the scaffold. One feature of the judged class deserves its own paragraph, because it returns in the findings and in the open problems. The rubric that sets the loss is itself a scaffold coordinate. Suppose a pitch-review system keeps overrating decks that claim a large market without numbers. It traces the failure to its rubric and edits "evaluate market size" into "require a numeric estimate and a source". That edit changes the gate that will judge the next edit. We call this the *reflexive gate*, and nothing in the corpus calibrates one: three scaffold systems report judge calibration [36], [44], [45], and none calibrates a judge that is itself being edited. The weight side faces the same question. Reinforcement learning from verifiable rewards lies outside the scored corpus, but pushing it into writing and other open-ended tasks has forced researchers to build judged gates. One group holds the gate fixed while the policy learns, as a checklist, a pairwise judge, a reference-anchored judge, or a fixed rubric [46]–[49]; a second group lets the gate move with the policy, scoring by the policy's own probability of a reference or updating the rubric or judge during training

TABLE 2. The four feedback-signal classes. Reading across a row: what the gate receives, what it can therefore be, and what it cannot fix.

Class	The signal the gate receives	What the gate can be	What stays hard	Corpus examples
C1 exact	A check that returns pass or fail, or an exact score, at once and repeatably: tests, compilers, answer keys	An automatic check; one run per case, repeated only to sample the task mix	The lift is bounded and specific, an improvement in what the check measures on the cases it ran; the check names the failed output, not the coordinate that caused it; a check that is also the target can be gamed	DGM, AlphaEvolve, SkillOpt
C2 observable outcome	A real result one episode later: a booked slot, a merged pull request, a closed ticket; noisy and confounded	An automatic threshold over batched episodes; a person at promotion	The outcome says *that* the episode failed, not *why*; the channel can be poisoned	Devin, Sierra, SAGE, SkillRL, PACE, Voyager
C3 judged	A rater, a rubric, or a model judge; no ground truth	A judge calibrated against people; a person at promotion	The rubric is itself a scaffold coordinate, so the gate judges edits to itself; the proxy gets optimized past the point where it tracks quality	SkillForge, CASCADE, skills-repository review
C4 delayed	An outcome weeks or months later, once per many edits	No per-edit gate; proposals are pruned and milestones stand in	Credit across many edits; drift; the loop cannot be told from random search	NanoResearch, CORAL, the AI Scientist

[50]–[54]. Fixed versus co-evolved is exactly the reflexive-gate question, and of the co-evolving methods only RLPR, whose reward is a probability rather than a rubric, reports agreement with human labels (Supplementary Material S6).

What the class decides. Read down the table and three consequences follow, each a prediction about the corpus that Section VII checks. First, which half of the loop can close automatically. An exact check or an observed outcome can gate a local edit by program. A judgment can do so only once the judge is calibrated, and a delayed reward cannot gate edits one at a time. Promotion is stricter. Only the exact class carries its own measure to the receiver, and only when the receiving side runs the same check: that is the one case in which the sending side's evidence means the same thing on arrival. In every other case the receiving-side evaluation of Section III is still owed. Second, which gate pairings a deployment can reach. [Auto $\times$ Auto] belongs to exact signals and simulated outcomes; [Auto $\times$ Human] belongs to observed outcomes and judgments, where the person on Loop 2 does the work the missing exact signal cannot; and the delayed class has no per-edit gate on either half. Third, which failure dominates, as the last column of the table records and Section X develops.

V. THE SURVEY: SUBSTRATES, CORPUS, AND SCORING

How the 256 systems were chosen and scored is set out here so that the trends of Section VI can be checked against the released table; a reader who takes the scoring on trust can go straight there.

A. SIX SUBSTRATES

A deployment's scaffold is composed of six substrates (Table 3). They are the coordinates of θ_s, and each can be edited independently of the others; the three kinds of memory on the memory substrate follow the cognitive-architecture taxonomy of Sumers et al. [55]. Other partitions are possible, for instance separating evaluation suites from governance. The argument of Section III is invariant to the partition so long as the coordinates stay independently editable. This partition is the one that recurs in practice: every integrated system in the corpus has a recognizable instance of each substrate under some name.

Every substrate is scored on the same 0 to 5 scale: 0, ephemeral; 1, persistent and local; 2, persistent with tool attachment but no feedback; 3, multi-scope or feedback-driven but without automated promotion; 4, automated feedback updates the scaffold, so one half of the loop is closed; 5, the substrate's own top mark in Table 3. Systems that adapt several substrates at once receive an additional integrated row. The top mark means different things on two kinds of substrate. On those that hold adaptable content (S1, S2, S5) it requires both halves of the loop. On those that hold governance machinery (S3 provenance and correction, S4 evaluation-gated tool versions, S6 evaluation-gated promotion with audit and rollback) it requires the machinery a second loop would need, not the loop itself. That is why tool registries and the prompt-management platforms of Section VI can score 5 with no inner loop at all and still sit outside the two-loop cells of the design space.

B. HOW THE CORPUS WAS ASSEMBLED

A system enters the corpus if it wraps a foundation model with at least one persistent artifact updated from deployment signal, or if it is described in public as self-improving, scaffolded, or agentic. The systems were found in public sources through the October 2026 cutoff and scored from their own method sections or documentation, never from a secondary

TABLE 3. The six scaffold substrates, what each holds, and what a system must do on it to score 5.

Substrate	What persists	A score of 5 requires
S1 Instructions	Project, team, or organization rules attached to a workspace	Versioned rules promoted under an automatic gate
S2 Skills	Procedures, as code or text, loaded on demand	A skill bank that edits itself on failure, test-gated, versioned, and promoted
S3 Memory	Experience kept across sessions: episodic, semantic, procedural	Provenance, versions, and a correction path for every entry
S4 Tools	The vetted action surface and its schemas	Evaluation-gated tool versions; a curated bundle as the product
S5 Orchestration	The multi-role graph and its routing	Topology updated from an outer-loop signal
S6 Governance	Versioning, promotion, audit, rollback	Staged promotion with evaluation thresholds, an audit log, and rollback

summary (Supplementary Material S2). Systems whose only update mechanism is weight-level training, and systems with no public evidence of deployment or release, are excluded.

Two filters structure the corpus (Fig. 3). The patch-local discriminator admits a system to the core corpus when it scores at least 3 on at least one substrate, that is, when some artifact persists across scope or is updated from feedback; systems that carry the labels but fail it form a contrast corpus, kept so that the boundary can be audited rather than asserted. Of the 256 systems, 204 form the core corpus and 52 the contrast corpus.

A stricter composite two-loop audit then asks two questions of each core system: does Loop 1 modify a persisted artifact from session signal, and does Loop 2 promote the artifact beyond the deployment that produced it, with versioning or lineage? Applied to the 204 core systems and to ten systems treated at mechanism level, which were classified from their papers and are not rows of the released table, it admits 14 research systems (13 table rows and SkillOpt) and 3 industry exemplars (Sierra Agent OS 2.0, Cognition Devin, and Letta's Context Repositories). Only these are placed in the design space. Passing the audit is a weaker thing than having the governed second loop of the thesis: every passing system promotes across users, teams, or repositories inside one operator, none reaches another organization, and one of them, Letta, promotes with no gate at all (Section IX).

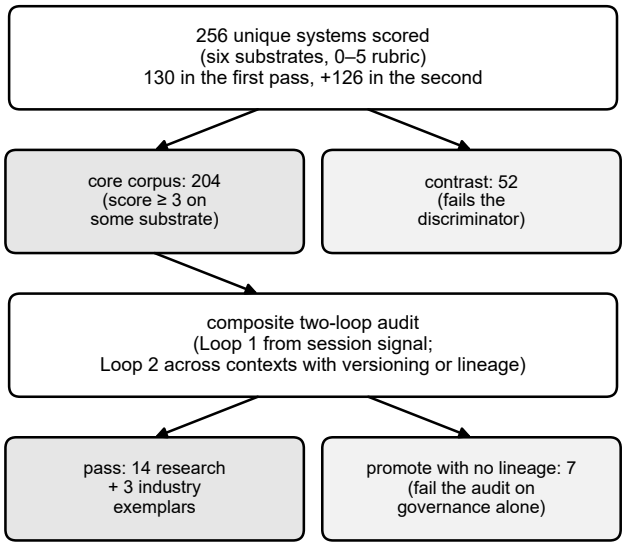


FIGURE 3. Corpus flow. Every system is scored on the substrates it adapts, with an integrated row for systems that adapt several, giving 269 substrate-level rows for 256 systems; the discriminator splits core from contrast; the composite two-loop audit selects the systems placed in the design space of Section VII.

C. EVIDENCE AND SCORING DISCIPLINE

Every classification rests on public evidence of uneven strength and on one rater. Each system carries one or more evidence tiers (T1 to T5 in the released table: peer-reviewed publication, arXiv preprint, production claim with metrics, open-source artifact, industry blog or interview). Peer-reviewed papers and open-source artifacts weigh most in cluster-level statements; a production claim counts when it names a benchmark or integration target; a blog post or interview counts only when a higher tier corroborates it; and production claims are never pooled with peer-reviewed evidence in a cluster-level statistic. Industry systems evaluated from vendor documentation rather than a paper are flagged wherever they appear. One author scored every system, used half-scores only when one criterion was clearly met and another partially, and rounded ambiguous cases down; the scores have not been independently re-rated, and that check is future work. The complete score table, with a one-line justification per row, is released so that every scored classification can be re-derived (Supplementary Material S1). The counts do move under natural relaxations of the composite audit (for instance, counting rollback alone as Loop-2 governance, asking for a promotion mechanism with no governance at all, or admitting a Loop 1 that persists only optimizer state); the qualitative findings survive every one of them (Supplementary Material S2).

VI. WHAT THE FIELD HAS BUILT

Two years of systems, read together, show three things. First, what changed: six developments, each visible in the released table, that together amount to most of the optimizer of Section III built one piece at a time. Second, what is still missing: five pieces of the second loop that no system supplies, each absent for a reason tied to its signal class. Third, how the field builds what it has: four kinds of system, each holding one half of the loop and none governing both. Between them, the three trace the gap the rest of the article is about: a first half that is routine, and a second half whose parts exist and have never been assembled.

A. SIX THINGS THAT CHANGED

The local loop is automated nearly everywhere. Nine in ten of the most recently added core systems close it under an automatic gate, and the pattern holds across substrates (Table 4). Skills are where the research activity is: SAGE and COSPLAY close both halves of the loop inside one research environment [33], [56]. SkillChain shows the local gate at its most careful, with judge-approved edits re-entering a human reflection gate before they reach the skill bank, a two-stage local gate rather than a promotion loop [44]. Orchestration followed the same path, from the automated workflow search of GPTSwarm's optimizable graphs [57] and AFlow's tree search over code-represented workflows [58] to topologies that evolve under evaluation [59]–[61], with TOCOMAS alone keeping a topology change only when it satisfies a structural invariant [62]. The counter-example is the most widespread artifact of all. Instruction files persist across IDEs and agents (Claude Code, Cursor, AGENTS.md [63], Windsurf, Continue.dev) and appear in a fifth to a quarter of coding-agent projects [64], [65], yet in the one controlled study they raise inference cost by about a fifth and buy nothing in task success [66]. A rule file that nobody measures is patchwork at scale, and it is the substrate on which no system closes both loops.

Evidence is batched before an edit is accepted. The earlier systems accepted an edit on one evaluation and absorbed the estimator's noise through the margin alone; the newer ones wait. SAGE-gate commits an edit only when a paired per-item test says its wins are reliable [67], and EvolveNet accepts a merge only when it fixes at least as many held-out items as it breaks [68]. In VaG, three heterogeneous critics must pass [69]; AutoSaddler feeds mini-batch failure signals into validation-based selection [70]. Two different operations hide under that one name. Requiring agreement among independent proposals drives the false-accept rate down geometrically; averaging over them shrinks the variance only linearly (Appendix XIII). Staged evaluation, by contrast, remains rare: AlphaEvolve is the only system that combines a cascade of evaluators with a diversity archive, and the other systems that keep archives use them for lineage, not for evaluation.

Retirement is gated. Eleven skill systems treat removing an artifact as an operation that must pass a check, the same as adding one [71]–[74]. The reason is in Ratchet's result: skills written by a language model gain nothing over a no-skill baseline until lifecycle hygiene is applied, while human-curated ones show a double-digit gain [75]. In other words, a pool nobody prunes starves the gate of signal before the loop can run (Section X). Hygiene is not an alternative to descent; it is what makes the gate readable.

Memory ships with its lineage, and correction is not enforced. Letta's git-backed Context Repositories [76], Mem0's per-memory history [77], and Zep's bitemporal facts [78] all record how an entry came to be, so a user who asks what the system remembers and why can be answered. Research memory splits into write-gating and provenance on one side [79]–[81] and verifier-gated failure loops on the other [82]–[84], and only Cuadros et al. does both, which is why it alone reaches the top mark. What lineage does not give is correction: five production backends still return facts that were revoked [85]. A memory that remembers its own history and cannot forget on request has the ledger of a second loop without its gate.

Formats and registries have converged. Claude Agent Skills, Cursor Rules, GitHub Copilot instructions, Windsurf, Continue.dev, Zed, and Replit all keep a git-tracked markdown artifact attached to a workspace, with an MCP-style tool layer beside it [65], [86]. In the public skill repositories, every substantive edit is human-authored and human-merged [87]. Registries govern what may be used, not whether it works. The MCP registries split into catalogues (namespaces and version records, no scanning) and control planes (scanning, access control, quarantine) [88]–[90], and the governance registries bring lineage and rollback [91]–[93]; none has evaluation-gated promotion, and none has an inner loop. Tools and governance therefore hold the machinery of a second loop without the first, which is how those substrates reach the top mark with no loop at all (Section V).

The first federations of deployments exist. EvolveNet evolves a shared harness across data-local clients and merges under a held-out gate [68]; Evo2Team promotes routing skills across teams and confirms them at the target [94]; a rule in XRepoSkill survives only if it is verified in at least two repositories [95]; and ShareMem consolidates one user's memories for another, then evaluates them there [96]. All four promote inside one operator. AlphaEvolve is the production proof that both gates can be algorithmic at once: an evaluator cascade paired with an evolutionary archive, as FunSearch did before it, with production gains at Google [30], [97].

No system matures all six substrates at once. Research systems learn fast and lack enterprise governance; production systems govern well and learn slowly; open standards solve artifact portability but not adaptation. This pattern is not a quirk of the corpus: it tracks which constraints each kind of system was built to satisfy, and it is the shape the next two subsections unpack.

TABLE 4. Substrate maturity across the 256-system corpus. Column n counts scored rows on each substrate, Mean is the average score, and Score-5 systems reach the substrate's top mark; the integrated row counts systems scored across several substrates as one composite row. Of the most recently added 126 systems, 31 are skill systems and 102 are core; 92 of the core systems close Loop 1 under an automatic gate, and none of the 126 scores 5 on any substrate.

Substrate	*n*	Mean	Score-5 systems	Survey finding
S1 Instructions	15	3.07	(none)	Rules persist across IDEs and agents; none closes both loops; ceiling 4
S2 Skills	50	3.34	SAGE, COSPLAY	Most active research substrate; Loop 1 routinely automated; production repositories human-merged
S3 Memory	50	3.06	Cuadros et al.	Lineage shipped in products; enforced correction still absent
S4 Tools	20	3.65	MCP, Harvey, Hippocratic AI	Most mature production substrate; registries govern admission, not evaluation
S5 Orchestration	37	3.08	(none)	Topologies evolve under evaluation; one structurally gated instance; no two-loop system
S6 Governance	45	3.02	Braintrust, Vellum, LangSmith Hub, AGENTS.md/AAIF	Mature prompts-as-code; registries add lineage and rollback; no inner loop
Integrated	52	3.81	NanoResearch, AutoAgent, SkillRL	High-water mark; research systems without enterprise governance

B. WHERE THE SECOND LOOP SHOULD BE, FIVE PIECES ARE MISSING

Walk the second half of the loop from the failure that starts it to the promotion that ends it, and five pieces are missing or rare across the core corpus. Each is missing in a particular feedback-signal class, which is the first sign that the gap is not one gap but several.

- **Failures are not captured** (C2). Deployed memory systems record preferences and successful facts, not residuals, so the loop has nothing to cluster. The one field report of stores written from dead-ends to prevent recurrence describes a single, self-reported deployment [98]; everywhere else the log that Loop 1 would cluster is never written.
- **Which sessions inform the next edit is left to chance** (C2). Nearly every system takes failures in the order they arrive. Task-CoEvolve's variance-weighted task sampling and AutoSaddler's mini-batch failure signals are the exceptions [70], [99], and both weight the sessions the loop learns from by how much they disagree.
- **Memory is versioned, but correction is not enforced** (C2, C3). Git-backed and bitemporal memory ship in products, and five production backends still return facts that were revoked [85]: the ledger exists, the gate that would act on it does not.
- **Promotion is rarely checked where the edit arrives** (C1, C2). ShareMem gives evaluation tasks to a different user, XRepoSkill verifies in at least two repositories, Evo2Team confirms at the target team, and VACE evaluates the candidate with the receiving model held fixed [100]; even Evo2Team's confirmation lets nearly a fifth of its transfers fail on new graphs. EvolveNet has no such check, and the federated optimization framework that proposes a receiving-client gate has not implemented it [101].
- **No promotion crosses an organization** (all classes). The four federations above, and RASO's adaptation of skills retrieved from a public corpus [102], promote across users, clients, teams, or repositories inside one operator [68], [95], [96]. No system promotes across customers or organizations with evidence.

The last piece is the thesis of this article. The four before it are the reasons a system that tried to add it would have little to promote and no way to check what it promoted.

C. FOUR WAYS THE FIELD BUILDS IT

Sort the core corpus by which half of the loop it automates and four clusters appear (Table 5). Three of them hold one half of the loop, the fourth holds both halves without governance, and none governs both.

Research full-auto systems show that the architecture works when both gates are algorithmic: NanoResearch, AutoAgent, FunSearch and AlphaEvolve, DGM, ADAS, and Voyager [29], [30], [41], [97], [103]–[105]. DGM is the textbook case: it proposes a modification to its own code, runs it in a sandbox against coding benchmarks, and keeps it only on a pass. The federations described above belong here too. What every member lacks is governance: no access control, no audited promotion, and no gate at a receiving organization.

Production hybrid systems show that the production-viable pairing is an algorithmic gate on Loop 1 with a reviewer on Loop 2. SkillForge illustrates it most plainly: a judge that agrees with people more than nine times in ten passes each skill diff, and support engineers decide whether it ships [36]. Sierra routes agent-generated insights through human-authored expert answers [106], Devin's merged pull requests are its promotion step [32], and Spotify launches automatically optimized prompts and plans only behind an online A/B test [28]. In every member, a person on Loop 2 stands in for the exact signal the cluster lacks.

Governance-first systems automate Loop 2 alone. The prompts-as-code platforms [107]–[109] and the governance

TABLE 5. Four clusters of the core corpus, by which half of the loop each automates.

Cluster	Which half is automated	What the cluster shows
Research full-auto	Both, under algorithmic gates	The architecture works end to end and benchmark gains compound; enterprise governance is absent
Production hybrid	The local half; a person decides promotion	The production-viable pairing: the reviewer stands in for the exact signal production lacks
Governance-first	Promotion only, under evaluation thresholds	Lineage, thresholds, and rollback exist as a product; nothing generates candidates from failures
Open standards and registries	Neither; people govern admission and access	The artifact format and the tool layer have converged across vendors; registries admit, they do not evaluate

registries [91]–[93] keep immutable versions, evaluation thresholds, and rollback, but they have no Loop 1: engineers author the candidates at a desk, an evaluation blocks a release on regression, and session feedback never triggers either. This is MLOps for scaffolds, distinct from MLOps for weights.

Open standards and registries hold the artifact and the tool layer without the loop. The convergence on a git-tracked markdown artifact with an MCP-style tool layer [65], [86] is what Section VIII reads as evidence that the architecture is real. Adaptation is what the cluster lacks: its registries govern admission and access but do not evaluate, and in the repositories that run autonomous loops, the runtime state of those loops sits outside version control [110].

VII. THE FOUR APPROACHES SIDE BY SIDE

The four clusters of Section VI differ in which half of the loop they automate, and the loop makes exactly two decisions: whether a local edit enters the scaffold, and whether an accepted edit leaves the deployment. Putting those two decisions on two axes gives the grid of Fig. 4. Each row is the gate on Loop 1 and each column the gate on Loop 2, with the three gate kinds of Section III; a cell such as [Auto × Human] holds the systems whose local gate is a program and whose promotion waits for a person. The grid is the comparison the clusters only hint at. It shows which combinations the field has built, which it has not, and whether the empty cells are empty by accident.

One thing jumps out of the grid: the field varies almost only along one axis. Every system with a gated local loop sits in the top row, and what tells them apart is the column, the gate on their promotion: a program, in the research full-auto cluster; a person, in the production hybrids; or nothing, in Letta's Context Repositories, which push every edit to shared memory with lineage and no check, and in the systems that promote nothing at all. Which column a system lands in is not a free design choice. Section IV predicted it from the signal: a program can gate promotion only on an exact signal or a simulated outcome, so [Auto × Auto] should hold exact-signal and simulated systems and nothing else; a person is the gate where the signal is an observed outcome or a judgment, so [Auto × Human] should hold those and no exact-signal system; and no system should calibrate a judge under edit. All three hold in the corpus. The grid therefore says something the

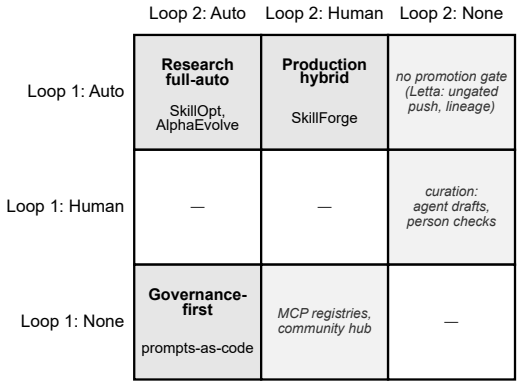


FIGURE 4. The two-loop design space. Rows give the Loop 1 gate and columns the Loop 2 gate, with None for a step that is absent or runs unchecked; a cell [Loop 1 × Loop 2] such as [Auto × Human] pairs an algorithmic local gate with a human promotion gate. Darker cells mark the three populated clusters; lighter cells hold curation, the registries, and the systems whose promotion is ungated or absent.

cluster table could not: the industry has converged on the local gate and is sorted on the promotion gate by the feedback it can get, which is the signal rule read off the systems themselves.

The empty cells and the lower rows say the rest. [Human × Auto] and [Human × Human] are empty: no surveyed system puts a person on the local gate of a failure-driven loop and then promotes the result, by program or by person. Only curation sits in the human-gated row, where an agent drafts and a person decides what is kept, as in VerificAgent's fact-checked and frozen memories [111] and SkillChain's reflection gate. The bottom row holds the systems with no failure-driven loop at all, which enter the grid as contrast entries (Supplementary Material S4). In [None × Auto] sit the governance-first platforms, which gate promotion on evaluation but generate no candidates; in [None × Human], the registries and community hubs in which a person registers an artifact and a person, or an admission scan, approves its use [88]–[90], [112]. One field study finds the same shape in practice: in its five public skill repositories every substantive edit is human-authored and human-merged, with one deprecation in 254 edits [87]. So the bottom row is where the governance

machinery lives and the top row is where the learning lives, and no system spans both under a gate on both axes.

Adding the seven systems that promote across users, teams, or repositories with no versioning or lineage would not change the picture; they would join cells already occupied, and none would open a new one (Supplementary Material S2). What no cell contains is governed cross-tenant promotion, and Section IX says what it would take.

VIII. FIVE RESEARCH FAMILIES CONVERGE ON THE LOOP

Five research lines, each started from a different problem, have built parts of the same loop without naming it, and we read them as evidence that the loop is where the constraints lead rather than one group's design. This section says what each family started from, which part of the loop it built, what it left open, and what their convergence does and does not show. The vendors' convergence on one artifact format and tool layer (Section VI) is independent evidence of the same kind. Convergence does not show that the design is optimal, only that the same constraints keep producing the same shape.

The clearest single instance is SkillOpt [31]. It treats one markdown skill document as the trainable external state of a frozen agent and edits it with explicit optimizer controls: rollout and reflection batches, add/delete/replace edits, a bounded textual learning rate with cosine decay, a held-out selection gate that accepts only a strictly better score, a rejected-edit buffer kept as negative feedback, and an epoch-wise slow update. Across six benchmarks, seven target models, and three harnesses it is best or tied on every cell its authors evaluate. It also reports transfer across models, harnesses, and benchmarks, which reads as an empirical measurement of evidence radius rather than as a governance mechanism. Its own follow-up, SkillOpt-Lite [113], removes the buffer, the slow update, and the schedule, keeps the held-out gate and a consensus filter across rollouts, and matches or beats it. So the gate and a consensus form of credit assignment are load-bearing; the rest, it turns out, are stabilizers (Supplementary Material S5).

The five families instantiate complementary parts of the loop (Table 6), and each was pushed into its part by the problem it started from. Reflective evolution (GEPA, and the fast-and-slow argument of Tiwari et al.) started from prompt optimization and continual learning: it proposes edits by reflecting over trajectories and argues on two timescales for keeping those edits outside the weights [4], [114]. Reflective context learning (RCL), the most direct independent formalization, started from the observation that credit assignment, overfitting, forgetting, and high-variance signals persist whether the learned object lies in parameter space or context space; its generalization penalty is the complexity term Ω of the gate [115]. Harness optimization (Meta-Harness, AHE, MOSS, CANTANTE, HARNESSEVO, HarnessForge, Growing Harness) started from orchestration rather than skills and names credit assignment as its bottleneck; Meta-Harness, for instance, keeps a filesystem of candidate harnesses that records which edit helped; that record is the accumulated history the loop reflects over. The family is also the largest in the corpus: roughly two hundred papers cite its founding entries, and a dozen are scored here [61], [116]–[121]. The lifecycle family (Ratchet, MOCHA, SkillsVote, Library Drift, SkillSmith, ASSAY) started from artifact pools that bloat and found hygiene to be a precondition for descent [75]: the complexity penalty becomes a constraint frontier once platform limits bind [122], each outcome is attributed to skill, exploration, or environment before an update is admitted [123], and eleven systems make retirement a gated operation [71]–[74]. The transfer family (ICT, Combee, CORAL, PACE, FederatedSkill, EvolveNet, Evo2Team, XRepoSkill) started from how far an edit travels and instruments Loop 2, from measuring evidence radius one task at a time to federating skill patches across clients or deployments; in FederatedSkill, a server agent applies, adapts, or rejects each client's patch and logs the decision [35], [42], [68], [124]–[126].

Read against the loop, the families support four conclusions.

1. The scaffold-versus-weights decomposition is independently confirmed. GEPA, Meta-Harness, MOSS, RCL, PACE, and the harness-optimization papers all begin from the observation that weights and the external artifact layer are distinct adaptation surfaces with different cost, reversibility, and scope, and a system that updates both keeps them as separate levers [127].
2. Credit assignment is the central engineering bottleneck. AHE, CANTANTE, Meta-Harness, HARNESSEVO, SkillsVote, and the harness-optimizer evaluation of Ong et al. [43] all name the same step as what separates descent from trial-and-error: attributing a system-level failure to one scaffold coordinate. That step remains unsolved in the general multi-substrate case.
3. Lifecycle hygiene is a precondition for descent, not an alternative to it. A dirty artifact pool starves the gate of signal (Table 8), gated retirement is the field's answer, and the one negative result cuts deeper: once a contaminated skill has descendants, rolling it back recovers little, so gating must happen before commit [69].
4. No family documents the governed second loop. FederatedSkill and EvolveNet each narrow the gap on different properties, and neither meets the four that Section IX sets out.

IX. THE MISSING ARCHITECTURE: GOVERNED CROSS-TENANT SCAFFOLD OPTIMIZATION

What is missing is specific. This section turns it into two things a reader can use: an audit criterion (four properties that any future system can be checked against) and a prescription (which of the four a deployment should expect to meet, given the feedback it has). Four properties define an architecture that no surveyed system instantiates:

- (a) **auto-gated Loop 1**: the commit criterion is an algorithmic threshold;
- (b) **cross-organization promotion**: Loop 2 routes artifacts between organizations (we use *cross-tenant* and *cross-*

TABLE 6. Five research families converging on artifact-layer descent: the problem each started from, the part of the loop it built, and what it leaves open.

Family	Started from	What it built	What it leaves open
Reflective evolution	Prompt optimization and continual learning	Proposal by reflection over trajectory evidence; the two-timescale case for editing outside the weights	A persistent artifact; a bounded step; a held-out gate
Reflective context	Learning in context as a setting of its own	The learned object in context space; a generalization penalty, the Ω of the gate	A cross-deployment Loop 2
Harness optimization	Orchestration and harness engineering	Credit assignment over the orchestration graph; source-level edits; slot attribution	A governed Loop 2; access control; rollback
Lifecycle and multi-objective	Skill pools that bloat under platform limits	Hygiene as a precondition; a constraint frontier as the shape of Ω; gated retirement	Closing the loop as descent; retiring a promoted artifact's descendants
Transfer and federation	How far an edit travels	Evidence radius measured one task at a time; cross-client aggregation; receiving-side confirmation	A cross-organization gate, lineage, and rollback

organizational interchangeably);

- (c) **versioned lineage with access control**: promoted artifacts carry versions, audit trails, and role-based access policies;
- (d) **rollback**: a bad promotion can be reverted without redeploy.

The closest approaches split these properties among them, and no row of Table 7 is full on promotion across organizations.

Row by row, the partition is tidy: the systems with a local loop carry the automatic gate and part of the promotion, the registries carry lineage and rollback without an inner loop, and the control planes carry access control alone. Rollback appears inside one federated round and inside single deployments, as edit-level reversal [72], [128] and as an admission test that rejects edits which cannot be undone [129]. No system rolls back a promotion that has crossed an organization boundary, because none makes such a promotion (Supplementary Material S4).

In machine-learning terms, what is missing is a cross-tenant aggregator with an automated evaluation gate at the scaffold layer. It would carry five parts: per-tenant artifact pools, each tagged with provenance; an aggregation step across tenants that respects tenant isolation; a held-out evaluation at the receiving tenant that blocks a promotion whenever it does not lower that tenant's patch loss; versioned lineage with access control; and a rollback envelope on every promoted artifact. The federated-learning analogy [130]–[132] is useful only in the limit where artifacts are mergeable update representations; federated prompt tuning, in which participants learn a shared prompt rather than shared weights [133], is the weight-side precedent closest to that limit. In surveyed practice Loop 2 is selection and merge: the nearest systems aggregate program edits with a language model, not by averaging.

Whether the architecture *should* be built depends on the feedback-signal class, not on the application domain (Section IV). Where the signal is exact, automatically gated cross-tenant promotion of a test-validated skill would be low-risk, and receiving-side evaluation is cheap to add. Where it is an observable outcome, the sending side's outcome does not transfer, so promotion should be automated only behind receiving-side evaluation, the check that re-establishes the lift under the receiver's own measure and would catch a scaffold over-fitted to one deployment before it propagates to another. Even that check is necessary rather than sufficient: some of Evo2Team's transfers still fail on new graphs after target-side confirmation [94]. Where it is judged, a human gate on Loop 2 is a design feature, and governance means lineage and rollback around a human decision. With a delayed signal, property (a) cannot be met; what remains is lineage and rollback without an automatic gate. Safety-critical domains, the usual case for human gates, are judged or delayed domains with high stakes. Whatever the class, the measure bound of Section III holds: where tenants measure different things, as they usually do across organizations, no lift transfers without the receiving-side check. Naming the four properties turns a vague gap into an audit criterion that any future system can be measured against; naming the signal class says which of the four a given deployment should expect to meet.

X. THE FAILURE SURFACE

While the second loop is missing, deployments pay for its absence in failures that weight-only systems do not have. They are the scaffold-layer entries in the technical-debt ledger that Sculley et al. opened for machine-learning systems a decade ago [134]. There are seven, each shown below in a measured case and paired with the piece of the optimizer that would close it (Table 8). Each dominates in one feedback-signal class, which is why the open problems of Section XI come one per class. The entries are documented attacks and degradations, not a new threat model.

Bloat is not only slow; it is a signal-quality problem. Instruction-following accuracy falls and latency grows sharply as the instruction count scales [135]. Ratchet reports what that does to the loop: LLM-authored skills gain nothing over a no-skill baseline until lifecycle hygiene is applied, human-curated ones show a double-digit gain, and the authors

TABLE 7. The closest approaches to governed cross-organization scaffold optimization, scored against the four properties.

System	(a) auto-gated Loop 1	(b) cross-organization promotion	(c) versioned lineage with access control	(d) rollback
FederatedSkill [126]	full	partial: privacy-preserving aggregation of skill patches across heterogeneous clients that sit under one operator	partial: auditable per-modification decision log	none
EvolveNet [68]	full	partial: held-out merge gate at the aggregator, but clients are benchmark partitions, not organizations	none: no lineage, no access control	full: rollback to the previous shared harness after a failed round
SkillForge [36]	full	partial: cross-team within one enterprise	partial	not documented
Governance-first registries [91]–[93]	none: no inner loop	none	partial: lineage without access control	full, with evidence
MCP control planes [88]	none: no inner loop	none	partial: access control without evaluation-gated promotion	not reported

TABLE 8. The failure surface of patchwork scaffold maintenance: what goes wrong, the optimizer piece that answers it, and the feedback-signal class in which it dominates.

Failure	What goes wrong	Architectural response	Dominant class
Scaffold bloat	Rules and skills accumulate until the model follows them worse and the gate has no clean signal	Complexity penalty Ω in the gate; pruning and expiry as first-class gated operations	C1, C3
Poisoned memory	An attacker writes an entry that later retrieval trusts	Provenance on every entry; write gating; a rollback envelope	C2
Injection via tools and MCP	A tool's output or a connected server carries instructions into context	Narrow tool permissions; a privilege hierarchy over instructions; tool bindings pass the same promotion gate as any other coordinate	C2
Evaluation fragility	The suite that gates edits is itself edited, gamed, and read non-independently	Held-out evaluations scoped within evidence radius, weighted toward recurring modes, re-read as the target drifts, checked against attribution	C3 (primary), C1
Coordination failures	Multi-agent roles blur and context is lost between agents	Orchestration topology treated as a scaffold coordinate under the same gate and rollback discipline	C2
Patch overfitting	A scaffold tuned to one deployment degrades another	Evidence-radius discipline at promotion; receiving-side held-out evaluation	C2
Plasticity loss	Old rules, polluted memory, and drifting tool versions crowd out adaptation	Expiry and rollback as explicit loop operations rather than retrospective hygiene	C4

attribute the gap to lifecycle management [75]. In the vocabulary of Section III, residual capture and mode discovery cannot work on a messy artifact pool, which is why gated retirement comes before descent, not after it.

Poisoning and injection are the two ways an adversary reaches the scaffold, and both arrive through the channel the observable-outcome class depends on. Memory is reached by writing: attacks against standard retrieval-augmented generation succeed at high rates [136], AgentPoison shows that backdoors can be planted in long-term memory or knowledge bases [137], and memory control-flow attacks work across the major agent frameworks [138]. Tools are reached by output: a tool's result or a connected server can smuggle instructions into context [139], [140], and one approved MCP configuration could later be modified without re-approval (CVE-2025-54136) [141]. The responses are the second loop's own machinery turned inward: provenance and write gating with a rollback envelope on memory [142], [143], and a privilege hierarchy over instructions [144] with tool bindings that pass the same promotion gate as every other coordinate.

Evaluation fragility compounds because the evaluation suite is itself a patch-local artifact rather than a fixed instrument. It is optimized against: once a metric becomes the target it is gamed, optimizing a proxy reward past a point lowers the true score [38], and access asymmetries alone produce large relative gains that reflect exposure to the test [145]. It is read non-independently, because the gate scores path-dependent agent trajectories whose readings do not compose like ordi-

nary test-set statistics. It goes stale, because the suite adapts on the same cadence as the scaffold it gates; and because failures concentrate in a finite catalogue of recurring modes, a suite that samples tasks at random under-weights exactly the families that drive the failure rate. The consequence has been measured: harness optimizers cannot be told from trial-and-error by downstream score alone [43]. From those facts the response follows: held-out and scoped, mode-weighted, re-read, and checked against attribution rather than only against score.

The remaining three are the price of structure. Multi-agent systems fail in fourteen measured modes, dominated by role confusion and context lost between agents [146], and the response is to treat the topology as one more scaffold coordinate under the same gate and rollback. A scaffold tuned to one deployment over-fits it by construction (Supplementary Material S3). Because the measure bounds the lift, the answer at promotion is evidence-radius discipline, with a held-out evaluation at the receiving side. And a scaffold that only grows loses plasticity the way a long context does: context rot [147], instruction conflict, memory pollution, and tool-version drift accumulate until expiry and rollback, run as loop operations rather than as retrospective hygiene, are the only way to recover it.

The failure surface is real, quantified in the cited measurements, and largely unmitigated in production. Every row names a piece of the optimizer that current systems do not yet have.

XI. OPEN PROBLEMS AND CONCLUSION

Five open problems follow from the survey, one per feedback-signal class and one for the architecture.

1. **Exact signals: credit assignment from a failing check to a scaffold coordinate.** A test tells the loop that an output failed, not which instruction, skill, or tool binding caused it. Slot attribution [120], intervention-based attribution [60], and contrastive multi-agent attribution [119] each solve one substrate's case; a general method is the bottleneck the harness-optimization family names.
2. **Observable outcomes: receiving-side evaluation before promotion.** The outcome that justified an edit on the sending side does not transfer, and even where a promoted edit is checked on arrival, some transfers still fail (Section VI). The gate a federated scaffold optimizer needs, a held-out test at the receiving tenant, has been proposed but not built [101].
3. **Judged signals: calibrating a gate that is itself being edited.** When the rubric is a scaffold coordinate, each accepted edit to it moves the instrument that judges the next edit. No method that co-evolves a rubric or judge reports agreement with human judgment, and no scaffold system calibrates a judge under edit (Section IV); the nearest measurement is on the human side, where evaluators' criteria drift as they grade [39].
4. **Delayed signals: telling descent from search.** Where the outcome arrives after many edits, an optimizer's gains cannot be distinguished from trial-and-error by outcome alone [43]; evaluating harness optimizers against their attribution claims rather than their scores is the open method.
5. **The governed cross-organization loop.** The architecture of Section IX, properties (a) through (d), is documented by no surveyed system; the receiving-side gate of problem 2 is the piece that would make cross-organization promotion safe in the two classes where it should be automatic.

The field has built an external adaptation layer for language models but not, as far as public evidence shows, the optimizer that should govern it. Both earlier papers of this series supply the hypothesis for why the layer belongs where it is: errors concentrate in a few key decisions; inside a bounded deployment, those decisions fail in a small and recurring catalogue of modes; and such a catalogue can be fixed one mode at a time in a persistent, inspectable, reversible artifact outside the weights. The corpus is consistent with that hypothesis. If it holds, the gated and scoped artifact layer is not a workaround but the architecture the error structure calls for.

Seen whole, the corpus says the industry has built most of that optimizer piecemeal. The local loop closes automatically in most research systems and in a growing number of production ones; evidence is batched before acceptance; retirement is gated; memory ships with its lineage; formats and registries have converged; and the first federations promote edits across deployments inside one operator. What nobody has done is assemble the pieces. No surveyed system promotes a validated edit across an organization boundary under an automatic gate with lineage, access control, and rollback. Read by signal class, this is not one gap but four, each with its own reachable answer. Under every class the same rule holds: a lift travels only as far as its measure does.

The frontier model generalizes. The local scaffold specializes. Reliability comes from governing that specialization. Because the instruments that judge each edit are themselves part of the moving scaffold, the same evidence-radius and lifecycle discipline the optimizer needs must extend to the gates that judge it.

XII. APPENDIX: FORMAL STATEMENT OF ARTIFACT-LAYER DESCENT

This appendix states the setup behind Section III, a one-step descent inequality under a noisy gate, an informal convergence proposition, the optimization-analogy correspondence, and four worked cases. No new theorem is claimed; the proposition is the ordinary target of local stochastic optimization under noisy estimators, restated for a discrete, versioned parameter.

A. SETUP

Let the frontier model have fixed parameters θ_M and let the scaffold be a structured object $\theta_s \in \Theta_s = \Theta_1 \times \cdots \times \Theta_6$ over the six substrate coordinates of Section V; Θ_s is discrete, symbolic, and versioned. For a deployment D, $L_D(\theta_s) = \mathbb{E}_{x\sim D}[\ell(\pi(\theta_M, \theta_s, x))]$. A candidate edit $z_t \sim Q_t(z \mid H_t, \theta_s^t)$

TABLE 9. Optimization concepts and their scaffold analogues.

Optimization concept	Scaffold analogue
Parameter vector	Instructions, skills, memories, tools, routing graphs, evaluation gates, governance rules
Loss	Failure, user correction, evaluation regression, retrieval miss, tool error, schema violation
Gradient estimate	Credit assignment identifying the coordinate responsible for a failure
Step	Accepted scaffold edit; step size is edit magnitude under $\oplus$
Aggregated update	Several recurring failures combined before an update, by averaging or by agreement across k proposals
Update radius	Promotion scope: which loss is descended (USER, PROJECT, STACK, TENANT, CORE)
Validation loss	Held-out patch evaluation before merge
Regularization	Complexity penalty $\Omega(z)$, bloat control, cross-patch regression checks
Overfitting	Scaffold improves on this deployment and degrades elsewhere
Rollback	Reverting a harmful accepted update

is drawn from a proposal process conditioned on the accumulated history H_t of failures, corrections, rejected outputs, evaluation regressions, and prior accepted edits. Q_t is not uniform: for a mode m that recurs with frequency f_m and carries per-occurrence loss s_m, its expected contribution to the failure rate is $f_m s_m$, so high-frequency or high-severity modes are proposed first. The finite directional difference is $\Delta_z L_D(\theta_s) = L_D(\theta_s \oplus z) - L_D(\theta_s)$; the gate sees only a noisy estimate $\widehat{\Delta}_z L_D$. The update is

$$\theta_s^{t+1} = \theta_s^t \oplus z_t \text{ if } G_t(z_t) = 1, \quad \theta_s^{t+1} = \theta_s^t \text{ otherwise.}$$

B. DESCENT INEQUALITY

Assume (A1) positive descent bias on accepted edits, $\mathbb{E}[\widehat{\Delta}_{z_t} L_D \mid G_t = 1, \theta_s^t] \le -\gamma_t - \tau_t$ for some $\gamma_t > 0$, and (A2) bounded estimator bias on accepted edits, $|\mathbb{E}[\Delta_{z_t} L_D - \widehat{\Delta}_{z_t} L_D \mid G_t = 1, \theta_s^t]| \le \varepsilon_t$. With $p_t = \Pr(G_t = 1 \mid \theta_s^t)$,

$$\mathbb{E}[L_D(\theta_s^{t+1}) \mid \theta_s^t] \le L_D(\theta_s^t) - p_t(\gamma_t + \tau_t) + p_t\,\varepsilon_t.$$

Per-step expected improvement is the gate's margin minus its estimator bias, weighted by acceptance probability. In the feedback-signal classes of Section IV, ε_t is near zero for the check itself in C1, with the sampling error of a finite suite over the patch remaining, bounded by batching for an observable outcome (C2), equal to the judge's bias and bounded only by calibration for a judged signal (C3), and unavailable per edit for a delayed signal (C4).

C. CONVERGENCE (INFORMAL)

Add (A3) proposal coverage with acceptance: at every non-gate-stable scaffold, Q_t places probability at least $q > 0$ on an admissible edit with $\mathbb{E}[\Delta_z L_D] \le -\delta$, and the gate accepts such an edit with probability at least $q' > 0$, so that $p_t(\gamma_t + \tau_t) \ge qq'\delta$ there; (A4) effective compactness of the reachable orbit; and (A5) summable bias, $\sum_t p_t \varepsilon_t < \infty$; and let $0 \le L_D \le B$. The one-step inequality is then of the Robbins–Siegmund form, a non-negative process that decreases in expectation up to a summable perturbation $p_t\varepsilon_t$, so $L_D(\theta_s^t)$ converges almost surely and $\sum_t p_t(\gamma_t + \tau_t) < \infty$ almost surely [148]. The finite sum carries the content: a bounded loss cannot pay for infinitely many accepted edits with a positive expected margin, so under (A3) the process visits non-gate-stable scaffolds only finitely often, and with (A4) the limit set of $\{\theta_s^t\}$ is gate-stable: no admissible edit clears the gate with negative expected directional loss in the limit. This is convergence of the loss and stability of the gate decision, not convergence to an optimum; a globally suboptimal gate-stable scaffold is consistent with the result, because Q_t may never propose the improving edit or $\widehat{\Delta}$ may be biased to reject it. In a non-stationary deployment ($D = D_t$) the same mechanism tracks a moving optimum rather than converging once, which is why recency weighting, pruning, expiry, and rollback are the scaffold-level analogues of continual-learning machinery rather than optional hygiene.

D. CORRESPONDENCE

Table 9 pairs each optimization concept with its scaffold analogue. Credit assignment here is attribution, not back-propagation. Scaffold systems do not differentiate through instruction files. They trace an output error to an upstream coordinate, manually or with a judge or verifier. The descent argument needs only that this attribution succeeds with non-zero probability per failure, which is the credit-assignment condition of Section III.

E. WORKED CASES

Retrieval. Failures recur because retrieved context omits the decisive clause. Attribution traces the error to the retrieval coordinate: chunking, query rewriting, metadata filter, embedding model, reranker, or context-packing rule. The candidate edit tightens chunk boundaries or adds a metadata constraint, and the gate checks that recall and answer accuracy improve on the patch evaluation without degrading adjacent cases. Signal class C1 or C2.

Pitch review. The system repeatedly overrates decks that claim a large market without numbers, and the failure cluster points at an under-specified rubric coordinate. The edit changes "evaluate market size" into "require a numeric estimate and a source; if absent, cap the score"; the gate runs the revised rubric against held-out decks with human scores.

What the edit changed was the gate that will judge the next edit: the reflexive gate of Section IV, class C3.

Tool use. Calls fail because optional parameters are underspecified. The responsible coordinate may be the tool schema, the call policy, or the clarification rule. The edit adds required fields or a pre-call uncertainty check; the gate checks that tool-call failures fall without raising refusals or latency. Class C2.

Memory. A stale customer preference is repeatedly retrieved and causes bad decisions. The responsible coordinate is a memory entry together with its provenance, priority, and expiry policy. The edit corrects, expires, or version-tags the entry; the gate checks that future behavior improves without erasing useful history. Class C2, with the correction itself observable only when a later outcome reveals it.

XIII. APPENDIX: GATE FAMILIES AND FEEDBACK-SIGNAL CLASSES

Section III collapses every production gate into one per-edit test G_t on an estimate of the loss change. Surveyed systems instantiate that test in several distinct ways, and some of them are better read as proposal pruning that precedes the per-edit gate than as estimators of $\widehat{\Delta}L_D$. Table 10 lists the families, the slot each occupies, representative systems, and the feedback-signal class (Section IV) each family serves.

The first five families populate G_t directly and, under the assumptions of Appendix XII, inherit its descent guarantee. Semantic merge and archive selection are not estimators of $\widehat{\Delta}L_D$; they constrain the proposal distribution, and the descent argument applies to the composition of pruning and gate. Multi-sample agreement is listed because it is the correct analogue of "wait for k pieces of evidence before accepting an edit", a different operation from mini-batch averaging and one that should not be conflated with it. Read down the signal-class column. The only families available to a judged (C3) deployment are the two whose estimator is a judge, and a delayed (C4) deployment has no family that estimates the loss change per edit at all: what it has is semantic merge before the gate and whatever proxy milestone stands at the horizon. That is the structural content of Section IV.

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TABLE 10. Gate families, their slot in the formalism, and the feedback-signal class each serves.

Gate family	What it estimates	Slot in Section III	Signal class	Representative systems
Surrogate verifier	Pass or fail from tests, a symbolic checker, or an exact match	Per-edit gate G_t	C1	DGM (sandboxed code), SkillOpt (held-out exact match), Voyager (game-state check: C2)
Evaluation cascade	$\widehat{\Delta}L_D$ on a held-out task suite, cheap-to-expensive evaluators	Per-edit gate G_t	C1 when the suite is exact; C2 when it is a task suite	AlphaEvolve, Braintrust, Vellum, LangSmith Hub
RL threshold	Reward improvement against a deployment threshold	Per-edit gate G_t	C2	SkillRL, SAGE, EvolveR, PACE
LLM judge with calibration	Pairwise or rubric score, calibrated against human agreement	Per-edit gate G_t	C3	SkillForge (Loop 1), CASCADE (consolidation gate)
Reviewer judgment	Human credit assignment by pull-request review or expert sign-off	Per-edit gate G_t, human-instantiated	C3 (and C2 when the reviewer has outcome data)	SkillForge (Loop 2), Sierra Agent OS 2.0, Anthropic skills repository
Semantic merge	Whether a candidate duplicates or conflicts with existing artifacts	Proposal pruning before G_t	C4 (also a pre-filter in C2, C3)	NanoResearch orchestrator, FederatedSkill server agent
Archive selection	Whether a candidate improves on, or diversifies, an archive of already-scored programs	Proposal pruning before G_t	Any; inherits the class of the evaluator that scored the archive (C1 in AlphaEvolve)	AlphaEvolve (MAP-Elites-style database), ADAS
Multi-sample agreement	Whether k independent proposals or evaluations agree; false-accept rate decays as p^k rather than variance as $1/k$	Aggregation filter on the estimate	C2, C3 noise absorption	VaG (three heterogeneous critics must all pass); SkillOpt-Lite (consensus filter across rollouts)

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